RAPID FRAMEWORK

VERSION 7.0

Published: September, 2025

Published at keyword: RAPID
Source in SharePoint site RAPID-Framework-Management (restricted access)

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Purpose of this Document

This document provides a description of the RAPID Compliance framework, version 7.0. It provides definitions of each RAPID work product including content guidelines and any specific compliance evidence required.

This document does not attempt to cover how to perform specific RAPID steps nor does it provide the underlying knowledge necessary to produce RAPID work products. This framework should be executed with a combination of inputs.

1. Industry knowledge and experience

2. RAPID specific information accessible via the RAPID website (kw: RAPID):

    - Job Aids
    - User Guides
    - Informational Sessions / Scheduled Office Hours

3. Use of the tools offered by the framework including support by one or more of the following:

    - Guidance provided in the tools for specific steps.
    - Guidance on DevSecOps tool usage can be found at Keyword: CICD Corner where you will find tool-specific information and a link to the Skills Hub where important articles explain the processes with recipes provided.

4. This RAPID Framework document

Abbreviations and Terms Used Throughout

The following abbreviations and terms are used throughout this document:

- **PDSM** – Process Driven Service Management built on the ServiceNow (SNOW) Platform
- **RDSO** – RAPID DevSecOps

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1 Definition of the RAPID Framework

RAPID is a set of work products and evidence standards focused on taking software development work from requirements to being move-to-production ready. Following this framework ensures that software developed at FedEx and for FedEx meets industry guidelines for quality software delivery and fulfills internal and external compliance requirements.

2 Applicability of the RAPID Framework

The RAPID framework applies to any changes to:

- FedEx developed applications
- Custom-developed or configured third-party software
- Applications hosted internally, in SaaS, CoLo, or cloud environments

3 Tools and Locations

ServiceNow Process Driven Service Management (PDSM) is the system of record for the software compliance requirements outlined in RAPID 7.0. All mandated evidence must be stored in PDSM.

The RAPID framework is published on a SharePoint site at Keyword: **RAPID**

The RAPID framework source material is maintained in SharePoint site RAPID-Framework-Management (restricted access).

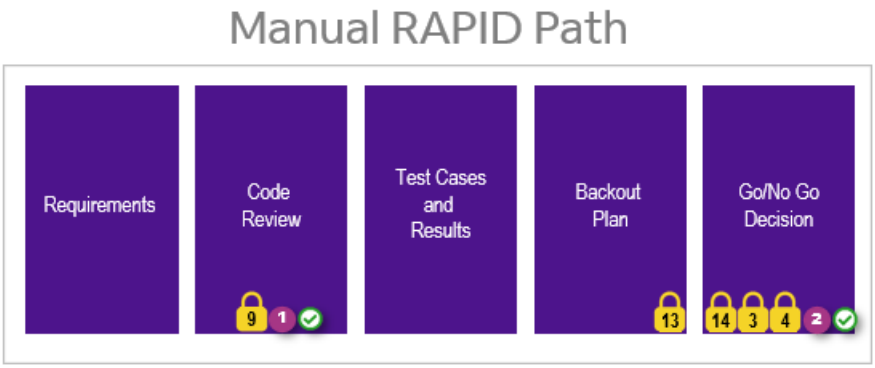
4 Related Processes and Standards

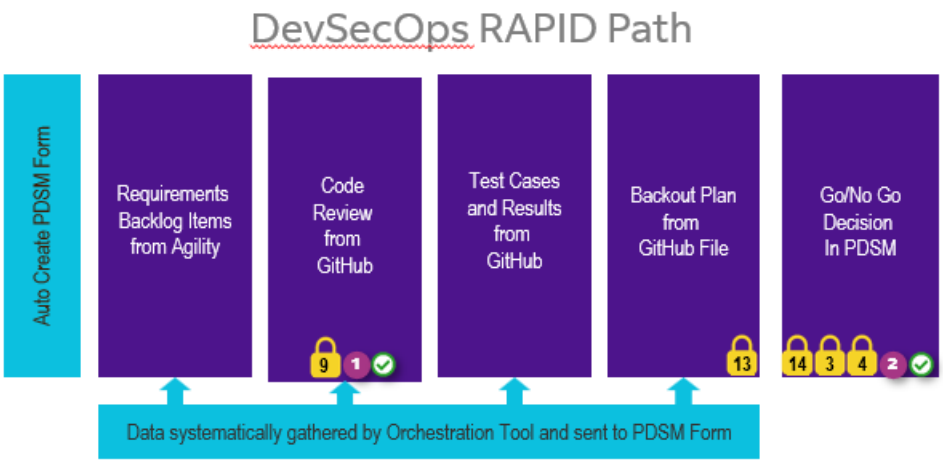
- **DRIVE Operating Model**
- **Common Standards**
  - SDLC Technical Standards
  - Agile Teams Ways of Working
- **SAFe** – Scaled Agile Framework
- **DevSecOps** – Development, Security, Operations (for those organizations adopting modern development practices)
- **QDM** – Quality Driven Management
- **ISC** – IT SOX Compliance

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5 RAPID Overview Diagrams

The diagrams below depict the work products and compliance steps to satisfy RAPID 7.0. It represents the framework that must be followed. RAPID 7.0 streamlines the RAPID Process and aligns to the new ITC change control numbers for both Manual (Legacy) and DevSecOps (Modern) RAPID paths. Subsequent sections explain the flow of each path.





Legend of Symbols :

- Dark Purple Rectangle – Work Product
- Purple Blend Circles with Numeric – Sequencing and Dependencies
- Green Circles with checkmark indicate steps requiring review and/or approval
- Locks – Represent Change Management controls. The locations on the diagram depict where a control is satisfied:
    - ITC.014 – Requirements
    - ITC.009 – Code Review and Approval
    - ITC.003 – Execute Test Cases
    - ITC.004– Go/No-Go Decisions
    - ITC.013– Backout Plan

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6 STEPS AND WORK PRODUCTS

RAPID consists of a defined sequence of steps and deliverables that must be completed. A work product represents a collection of one or more items that satisfy a deliverable. The sections below provide more detail on each step and outline the guidance to follow for each required work product.

6.1 CONSIDERATIONS BEYOND RAPID STEPS AND WORK PRODUCTS

There are activities necessary to produce working software that are not described as part of the RAPID framework. While they are not detailed here, these steps rely on industry knowledge and general software development practices to produce a successful outcome. The following are examples of these types of activities:

- Evaluating that the work to complete is desirable (ROI, etc...)
- Managing scope
- Stakeholder analysis (reviewers and approvers beyond the required minimums)
- Software evaluation and selection (RFI, RFP, etc...)
- Storing software in a configuration management system
- Building software via a repeatable, automated system
- Deploying software via a controlled, automated system
- Automating tests
- Managing defects
- Planning for deployment
- Post-deployment software checkout

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7 REQUIREMENTS WORK PRODUCT

7.1 DEFINITION

Requirements are desired outcomes, business rules, and an answer to the question: "what does the solution need to do?"

- Expressed as Stakeholder Needs
- And Desired Solution Capabilities

Requirements are also what the software must do and how it must perform to meet the desired outcomes.

- Software Behavior
- Software Interaction

7.2 PURPOSE

Requirements serve as a foundation for further analysis, are referenced when building test cases and used to inform the coding activity. Thorough Requirements ensure that all involved understand what is to be delivered, and that the need will be met, with clear boundaries of what is to be created and delivered.

7.3 APPLICABILITY

Requirements are required whenever changes are made to a FedEx owned application to deliver value to the application's end users and customers. This includes 3rd Party software for which FedEx does not have source code access, but which is modified at the direction of FedEx, and those modifications are released only to FedEx (not part of the 3rd Party's Generally Available (GA) version).

Requirements must exist for all software changes.

7.4 CONTENT GUIDELINES

- Agile teams should follow the common standards and ways of working outlined by the Enterprise Business Agility Frameworks to deliver tech-enabled value to the organization.
- Requirements can be described to varying levels of detail. Choose the level of detail that allows for the best understanding between business stakeholders, the product owner and solution delivery team.
- Requirements can take many forms, for example, they can be written statements, pictures, business process flows, diagrams, use cases, user stories, feature descriptions, tests, defects, acceptance criteria or output generated from an analysis model. Choose the appropriate capture and communication methods to gain the best understanding between the business stakeholders, the product owner and solution delivery team.
- Requirements can be described from many perspectives, for example, from a feature, business, application, system, software, transitional, functional and/or non-functional perspective. All perspectives are valid. Choose the appropriate perspective(s) to gain the best understanding between the business stakeholders, the product owner and solution delivery team.
- Requirements can cover many subjects in defining business needs and solution capabilities. For example: Performance, Security, Usability, Reliability, Legal & Regulatory, Maintainability, Configuration, Languages, Localization, Supportability, Business Rules, Software Behavior, Data Validation, User interfaces, Screens,

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Reports, Error Handling, Decoupling and Computations. Ensure all subjects are covered that are necessary to represent the stakeholder needs and solution capabilities.

7.5 Storage

Requirements can be authored within Agility as user stories or defects, within the ValueEdge Requirements or Defect modules to which they are most closely related, or within the PDSM RAPID Controls and Go/No Go Release Requirements Task. Requirements may be authored in other locations, such as Word, PowerPoint, Excel (Office 365, etc.). The version of Requirements which will be used for Requirements evidence must be accessible from PDSM. This may require the end user to extract those Requirements from the authoring tool and upload them to PDSM.

Each Requirement must have a unique identifier (typically a number either system generated or manually created. The Agility backlog ID is an example of a system generated identifier).

7.6 Review Acceptance Criteria and Quality Standards

- Each Requirement should be understandable by the business stakeholders, the product owner and the solution delivery team.
- Each Requirement should be at an appropriate level of detail to define stakeholders' needs and the solution capabilities such that the stakeholders, product owner and solution delivery team can work together to create the solution.
- Each Requirement should be clear and accurate.
- Requirements should not overlap with others in the collection.
- Each Requirement should be testable.
- The Requirements must address Information Security standards as either part of this solution set or as part of an existing set of Requirements from another solution set.
    - InfoSec Access Control Standards such as authentication, authorization of resources and consideration for a user's need to create, read, update and delete data.
    - Controls to protect data classified Sensitive or Internal per InfoSec Standards.
    - Audit logging in accordance with Information Security Standards.
    - Data encryption and controls in accordance with InfoSec Standards.
    - Data Encryption or Obfuscation for data classified as Sensitive per Information Security Standards.

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8 Code Work Product

8.1 Definition

Documents the software that is created or changed to get the desired outcomes.

8.2 Purpose

Captures the evidence that a Code Review occurred which meets the minimums described in Review Acceptance Criteria and Quality Standards.

8.3 Applicability

The Code work product is required for each RAPID software development release in PDSM.

8.4 Storage

- Code must be stored in an approved source control environment (E.G., Enterprise GitHub).
- Evidence of a Code Review must be stored in PDSM

8.5 Review Acceptance Criteria and Quality Standards

- Sufficient information is provided as to what Code has been modified (files, versions, etc.).
- Sufficient information is provided to know who modified the Code and who reviewed the Code (must be at least one qualified peer other than who made the modification).
- Peer reviewer agrees the Code meets Information Security guidelines.
- Peer reviewer agrees the Code has been written using Reference Architecture guidelines (or can support why it wasn't in the cases that it isn't).
- Peer reviewer agrees the Code is properly documented based on functional area standards.
- Peer reviewer agrees the Code will perform only the actions required (as stated or understood from the Requirements).

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10 Test Cases & Results Work Product

10.1 Definition

Validation to ensure individual software components/modules work as designed and/or to ensure the software has the capabilities to solve the stakeholder needs and the required behaviors and interactions are present.

10.2 Purpose

To ensure that every Code change is tested including component level checking of every point of decision logic.

To ensure that the change satisfies the Requirements.

10.3 Applicability

All Code changes must be tested.

10.4 Storage

- Test Cases can be authored in several tools, most notably written along with the software being modified and stored with the source code. Additionally, Test Cases can be authored in other tools (Agility, ValueEdge, Office products, etc.).
- Test Results should be captured from automated tools wherever possible or alternately documented in other tools (Agility, ValueEdge, Office products, etc.).
- The Test Cases and Results (or a summary as described in content guidelines) from a significant test run (I.E., last/latest run) should be captured and stored in PDSM for compliance evidence.

10.4.1 Additional Information

Additional information can be provided in the Detailed Description field of the Test Cases and Results task such as:

- The type of test(s) performed and/or pass/fail metrics. Test types can include Unit, System, Integration, Regression, User Acceptance, Vulnerability/Security, Usability, Performance, Load Stress, Localization, Decoupling, Destructive.

NOTE: Per InfoSec policy, it is imperative that no live sensitive data be used in the conduct of testing. If testing requires manipulation of sensitive data fields, then mock data will be generated for that specific purpose.

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10.5 Content Guidelines

- The Test Cases & Results work product must contain the actual test cases and the results of those tests (pass/fail).
- If each of the Test Cases are being executed and the results are being tracked via a tool (For example: ValueEdge TEST/DEFECTS, and the Test Cases and Results are impractical to extract and store in PDSM, then a summary of the Test Cases and Results can be provided.
    - When providing a summary of Test Cases and Results:
        - The location(s) to the actual Test Cases and actual Test Results must be provided.
        - The number of Passed, Failed, Not Able to Validate (including reasons why), Remaining (including reasons why) test cases.
        - Any outstanding defects and their severity

10.6 Quality Standards

Captured Test Cases and the Results or a summary of the Test Cases and Results.

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11 Backout Plan Step

11.1 Definition

The Backout Plan specifies the steps required to recover to a known working state if a release fails in production.

11.2 Purpose

The purpose of the Backout Plan is to document the strategy for returning to a known working state in production, if a software release or portion of a software release must be backed out.

11.3 Applicability

Each release to production (PDSM RAPID Controls and Go/No Go Release) must have a documented Backout Plan.

11.4 Storage

The Backout Plan must be provided or referenced on the PDSM release. If a document is referenced, it must be uploaded/stored on the release, release phase or release task.

Note: For DevSecOps, the Backout Plan is pulled from GitHub and stored as an attachment at the Release level.

11.5 Content Guidelines

The Backout Plan must contain whatever information would be needed for someone else to back out the change. The plan should be readily available to reduce any business impacts.

- If files need to be manually restored, specify the locations of those files. Simply stating “Restore previous version” is not sufficient. Locations and names of the files must be specified.
- If an automated back out is to be performed using custom scripts, provide commands that should be executed with any required parameters.
- If an automated move tool is utilized, provide all package IDs and version information that the system will require to perform the back out.
- If the back out of a change has cross impacts with other changes or systems, document those along with what steps need to be taken to ensure the integrity of the cross impacted systems, along with contact information.
- If the release cannot be backed out, the Backout Plan must clearly state this and contain a contingency plan explaining how to address any issues that arise after the move to production.

11.6 Quality Standards

The Backout Plan must address any steps required to recover for this release.

- The Backout Plan must contain enough information so that either an individual not familiar with the release can accomplish the backout or contain names and contact numbers for those that can. If the actual Backout Plan is not documented on the Go/No-Go, then there must be a document with the plan uploaded to the Go/No-Go in the field provided.
- Notations (alone) such as “roll back to previous version” or “see documented Backout Plan” are insufficient.

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12 Go/No-Go Approval

12.1 Definition

Management is responsible for accepting the risk of each change to a FedEx application. The Go/No-Go represents an agreement between both business and IT management, or the Application IT Owner and the Application Business Owner, that the software included in the release meets expectations and can be released into the production environment.

12.2 Purpose

The Go/No-Go provides a means to capture the agreement from the Application IT Owner and Application Business Owner and an acknowledgement that the software meets RAPID standards and can be released to production.

12.3 Applicability

Each release must be approved. Only appropriate members of management can approve changes to FedEx applications. Prior to moving changes to production, a "Go/No-Go" decision must be made by IT and user management, documented, and retained. Application Business Owner and IT Owner must be valid employees and a member of FedEx management as listed in the LDAP hierarchy.

12.4 Storage

The Go/No-Go must be completed using the RAPID Controls and Go/No-Go Release template form in PDSM. Approver decisions are time and date stamped in the Go/No-Go decision task within PDSM Release and retained.

12.5 Content Guidelines

The Go/No-Go decision task in PDSM requires management's attestation that RAPID standards, and embedded compliance controls, have been followed and that risk has been reviewed and accepted before moving changes to production.

The attestation statements agreed to by the Application IT Owner and the Application Business Owner in PDSM read as follows in both the Legacy and Modern development paths:

- For IT Owners:
    - "As IT Owner (or acting delegate for) [IT Owner Firstname LastName (FedExID)]: *In accordance with application change management compliance requirements, IT Owner Go/No-Go approval of requested changes to [Application ID:Application Name, RLESE# Release Name] represents that you are the appropriate member of management to approve changes to this application, have evaluated whether all functionality changes satisfy the requirements, coding was independently reviewed, as applicable, in accordance with FedEx secure coding guidelines, sufficient testing has been conducted including all required application security testing (keyword: securitytesting), all business risks have been reviewed and accepted (including acceptance of all unresolved defects), backout plans have been documented to allow-timely removal of code, if warranted, following the move to production and that you approve the code changes to be moved into the production environment*"

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- For Business Owners:
    - "As Business Owner (or acting delegate for) [Business Owner Firstname LastName (FedExID)]: *In accordance with application change management compliance requirements, Business Owner Go/No-Go approval of requested changes to [Application ID:Application Name, RLESE# Release Name] represents that you are the appropriate member of management to approve changes to this application, have evaluated whether all functionality changes have been documented, sufficient testing has been conducted to ensure functionality changes satisfy the documented requirements, all business risks have been reviewed and accepted (including acceptance of all unresolved defects) and that you approve the code changes to be moved into the production environment. In some cases, the aforementioned changes will result in no impact to the business functionality of this application, however your approval is still required as the primary business owner of this application."*

12.6 Quality Standards

Each work product in the release has met individual work product standards. Reviews as required by the RAPID framework have been completed.

For each RAPID work product that is not formally reviewed, as allowed per RAPID rules, the Application IT Owner and Application Business Owner agree the work product still meets minimum quality and compliance standards.

12.7 Periodic Assessment

Application Business Owner and IT Owners are validated on a periodic basis to ensure appropriate members of management approve changes to FedEx applications. When a manager approves changes to production, they indicate they are the appropriate person to approve said changes. Management is responsible for ensuring application ownership remains current and accurate as organizational changes occur.

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13 Definitions of Evidence

Each control requires documented evidence that can be easily located. Evidence must be retained and static in electronic form and must prove the control has been satisfied.

13.1 Evidence Best Practices

- Stands on its own merits and clearly communicates its intent (E.G., a third party can look at it 10 months later and have no questions as to what the evidence is stating).
- Must be complete, informative, and pertinent to the software change it supports (E.G., small for one-line Code changes – significant for a system re-write).
- Should answer who, what, when, where, why, and how.

13.2 Practices to Avoid

- Commentary such as "I walked through my code," without documentation. If evidence does not exist, an auditor must assume the task did not happen.
- Control was performed and documentation was created, but documentation cannot be located. If evidence cannot be located, an auditor must assume the task did not happen.
- Does not follow logical sequence (E.G., Requirements approved after software was loaded to production).

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14 Emergency Changes to Production

14.1 Definition

An emergency change to production is one that must occur immediately to restore an application to previously defined and expected functionality. Emergency changes to production can be a result of an incident in PDSM or coding and testing activities that bring the emergency situation to light. An emergency change to production is only authorized to restore production functionality (assuming the functionality had already been approved by the business).

- **An emergency change can only restore current functionality.**
- **An emergency change can never add to or enhance the current production functionality.**
- **It will correct a loss or prevent a potential loss of any function and/or data in production, which puts any critical business operation/function at risk.**

14.2 Process Execution During Emergency

14.2.1 Manual

In an emergency situation, the coding and testing of the fix must still be conducted in a manner that reduces production risk to acceptable levels prior to placing the fix in production. It is only the compliance documentation that is temporarily bypassed for operational expediency.

14.2.2 DevSecOps

In an emergency situation, the coding and testing of the fix must still be conducted in a manner that reduces production risk to acceptable levels prior to placing the fix in production. It is only the SDLC Compliance documentation of the Go/No-Go that can be temporarily bypassed with the correct authorization in the Application Release Orchestration Tool (see the Documentation Section for additional required steps).

14.3 Documentation of Emergency Change

SDLC Compliance evidence and RAPID-mandated approvals must be completed by the end of the next business day after production implementation (i.e. an emergency change identified and corrected on Friday night would need to have all compliance controls documented by end-of-day Monday. If Monday is a holiday, compliance controls must be documented by end-of-day Tuesday). This includes the following:

14.3.1 Manual

- All required work product actions, reviews and valid RAPID Controls and Go/No-Go Release will be completed in the system of record (PDSM) within one business day of the emergency change to production.

14.3.2 DevSecOps

- Ensure the RAPID Controls and Go/No-Go Release is created and the decision package from the Enterprise CI/CD pipeline is presented to the Application Business Owner and IT Owner (or their delegates) within one business day of the emergency change to production.
- Ensure the Application Business Owner and IT Owner provide their decision on the RAPID Controls and Go/No-Go Release within one business day of the emergency change to production.

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15 Process Change management, Governance and Requests for Exceptions

15.1 Changes and Governance

Formal changes to this process framework will abide by the following:

- Drafted changes for publication will be submitted for review and formal approval to both IT SOx Compliance and Financial Compliance.
- Additional processes are required to be followed for an exception to RAPID. See standards and controls procedures for the enterprise for additional information.

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16 REVISION HISTORY

Date of Revision	Version	Author	Revision Notes
9/29/2014	v.04	J. Pyatt	Initial Draft
10/8/2014	v.05	J. Pyatt	Incorporated feedback from internal review
10/24/2014	v.06	J. Pyatt	Incorporated initial feedback from audit stakeholders Reordered work product and compliance sections and added purpose of document section.
11/18/2014	v.07-.08	J. Pyatt	Minor wording modifications based on informal feedback
12/30/2014	v.09	J. Pyatt	Differentiated more clearly between "flows/process flows" for SHLD and Requirements. Requirements have business process flows, SHLD have system process flows (when these need to be differentiated they are more appropriate for each of these work products)
8/31/2015	V2	J. Pyatt	Revised formatting throughout. Modified process flows to fulfill requests for sequencing and rule changes. Added sections for Governance and Methodology guidance. Changed SHLD section to address needs of EADS (now Architecture Delivery); incorporated SAS to DES conversion wording. Added Definitions of Evidence Section (source: Compliance Guide for Using GDP) Added Two Managers' Approval definition section Added Transition to RAPID section Added Considerations for Steps outside of RAPID Added Defect Management Note
1/5/2017	V3	M. Rabinowitz	Removed the GDP-to-RAPID transition section Removed the Process exception section Added PPM references Added Opt Out section Added new content for Problem Statement, Solution High Level Design, Backout Plan, Implementation Plan and Test Plan work product sections. Added Process Governance section
1/12/2017	V3	L. Prestage-Mosby	Proofed and formatted
2/6/2017	V3 FINAL	M. Rabinowitz	Final updates from review cycle
5/5/2017	V4	M. Rabinowitz	Added Emergency Change section, updates to SHLD work product
5/11/2017	V4.1	M. Rabinowitz	Changed Emergency Change section to remove references to the PDSM system.
2/10/2018	V5.1draft2	M. Rabinowitz	PPM and ALM-specific updates.
1/12/2020	V6 Draft	J. Pyatt	Made revisions throughout. • Added Cultural interactions with RAPID • Removed Defect Management section as it reflected a waterfall perspective. Managing defects is still listed in "beyond RAPID" section

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			• Added wording for Modern, DevSecOps path • Split where applicable to Legacy and Modern • Updated images throughout based on RAPID on a Page Diagram • Removed references to AGM, Added reference to VersionOne • Reduced ALM references and changed wording to reflect current expectations • Added Modern Test Cases & Results section • Removed Tools and Locations section (redundant with other content) • Added additional clarifications to Go/No-Go Decision section • Added clarifications to Two Managers' Approval section • Renamed Emergency Production Problem section to Emergency Changes to Production to reflect DevSecOps scenarios. • Refined wording in Emergency Changes to Production to reflect DevSecOps scenarios. • Refined Grammar and formatting throughout
11/27/2023 – 1/31/2024	V6.5 Draft	L.A. Kutil	Made revisions throughout for RAPID 6.5 and RAPID 6.5 for LPM. • Added information about the Initiative and Epic as a starting point for the RAPID 6.5 for LPM paths. • Replaced diagrams for the overall process and individual RAPID paths. • Added Paths for LPM for both Legacy and Modern. • Updated references to the Enterprise Code Repository, GitHub and ability to user ephemeral testing environments. • Clarified the timeline for documentation for emergency changes to Production to be one business day after the production implementation. • Refined Grammar and formatting throughout.
Jan 2025 – 9/15/2025	V7.0 Draft	L.A. Kutil	Made revisions throughout for RAPID 7.0 and change of implementation tooling.

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